1 Evan Roth, *Since You Were Born*, 2019 (detail)

World of Art

Digital Art

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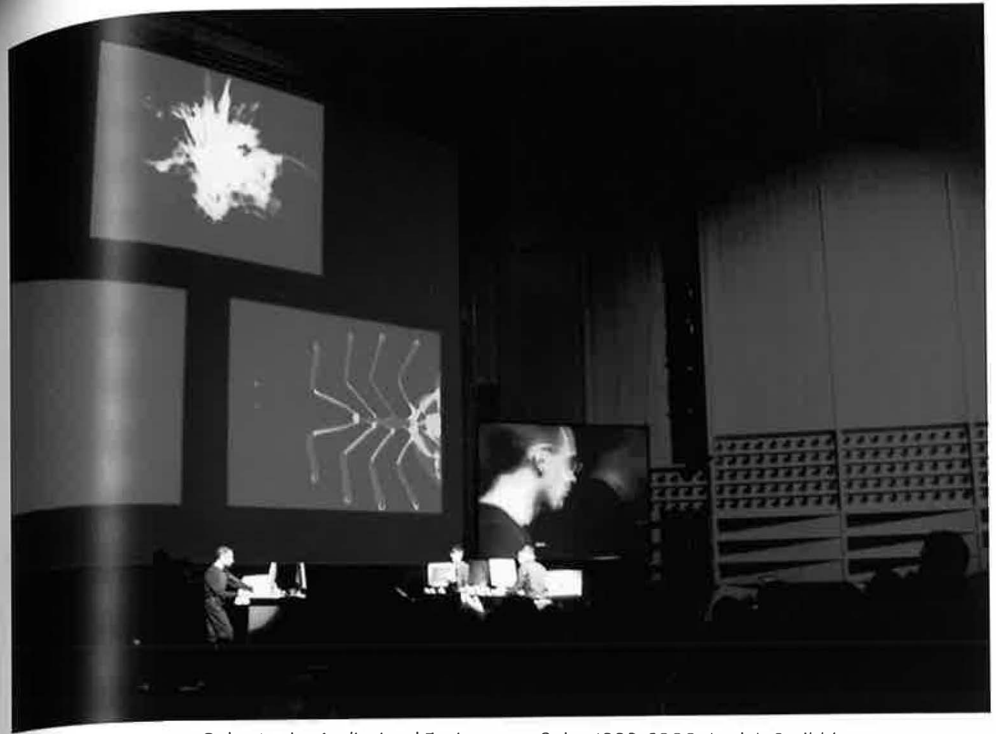
Fourth edition

Sound and Music

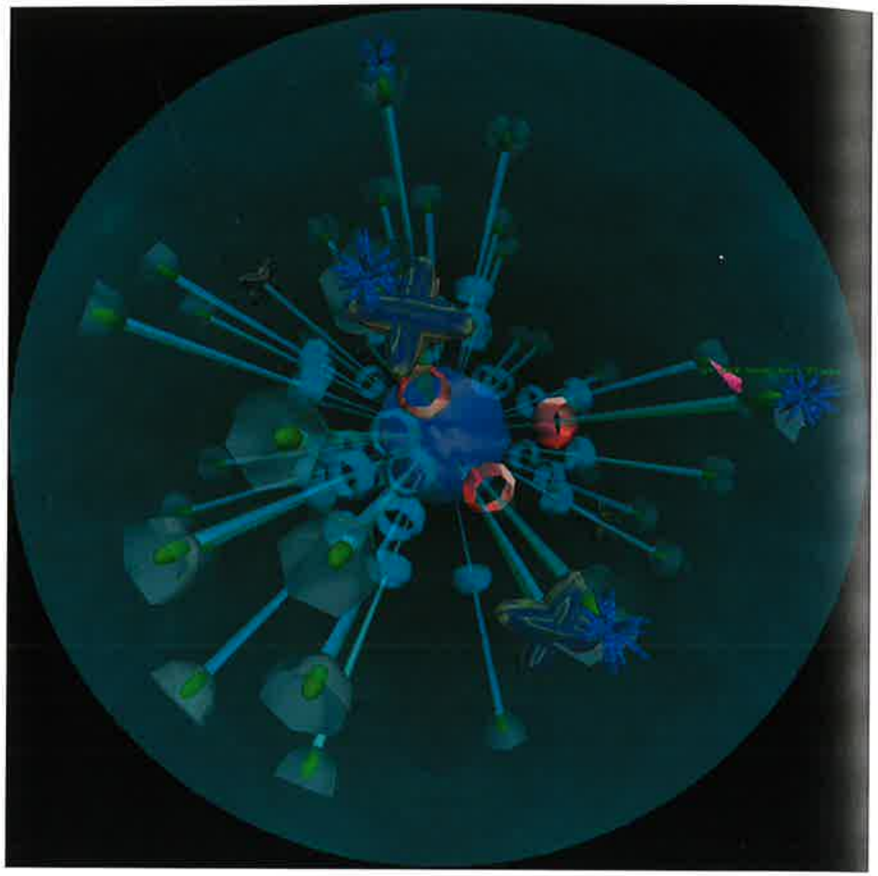
The impact of digital technologies on the creation, production, and distribution of sound and music projects is a vast topic that would require a book of its own and can only be briefly outlined here. Computer-generated music was in many ways a driving force in the evolution of digital technologies and concepts of interactivity itself, and is also closely connected to the history of electronic music.

Besides technological developments, the evolution of digital sound and music was shaped by a multitude of earlier musical experiments that pointed to the possibilities of the new medium. John Cage's work with found sounds and rules, or Pierre Schaeffer's *musique concrète*—a term Schaeffer coined in 1948 for composing with materials from an existing collection of experimental sounds—are highly relevant to the digital medium's possibilities of copying and remixing existing music files. Brian Eno's sound environments and Laurie Anderson's audio-visual installations/performances also had a profound influence on developments in digital sound and music. Apart from remix and DJ culture, artistic digital sound and music projects are a large territory that includes pure sound art (without any visual component), audio-visual installation environments and software, Internet-based projects that allow for real-time, multi-user compositions and remixes, as well as networked projects that involve public places or are distributed as apps on mobile devices. Many digital artworks, from installation to Internet art, involve sound components without being specifically focused on musical aspects. The projects described below are just a few examples of the use of digital technologies in the field of sound and music.



129 Golan Levin, *Audiovisual Environment Suite*, 1998–2000. *Levin's Scribble* (2000), a concert performed live with the Audiovisual Environment Suite software, created an interplay of scored and improvised composition, and examined the possibilities of experimental interfaces for a multimedia performance. The *Suite* software consists of five interfaces for audio-visual composition that allow performers to create abstract visual forms accompanied by sounds. Each of the interfaces differs in its relationships between visuals, the pitch and tone of sounds, and the performer's movement of the mouse.

Among the artists who have explored communication protocols in the simultaneous creation and interconnection of image and sound is Golan Levin (b. 1972), an artist, composer, performer, and engineer who received his degrees from the MIT Media Lab, where he studied with John Maeda in the Aesthetics and Computation Group. Levin's *Audiovisual Environment*
129 *Suite* (1998–2000), an interactive software that allows for the creation and manipulation of simultaneous visuals and sound in real time, strives to establish inherent, 'organic', and fluid connections between the unfolding of musical and visual form. Many of the digital media projects focusing on the combination of visuals and sound stand in the tradition of kinetic light performance or the 'visual music' of the German abstract animator and painter Oskar Fischinger (1900–67).



130 John Klima, *Glasbead*, 1999

The concepts of multi-user environments, gaming, and file-
sharing are central to John Klima's (b. 1965) software *Glasbead*
130 (1999), a multi-user collaborative musical interface, instrument
and 'toy' that allows players to import sound files and create
a myriad of soundscapes. The interface consists of a rotating,
circular structure with stems that resemble hammers and bells.
Sound files can be imported into the bells and are triggered by
flinging the hammers into the bells. While *Glasbead* creates
a contained world where sounds and visuals enhance each
other, it allows up to twenty players to remotely 'jam' with each
other. The project was inspired by Hermann Hesse's novel *Das
Glasperlenspiel* (*The Glass Bead Game*, also published in English
under the title *Magister Ludi*), which applies the geometries of
music to the construction of synaesthetic microworlds.

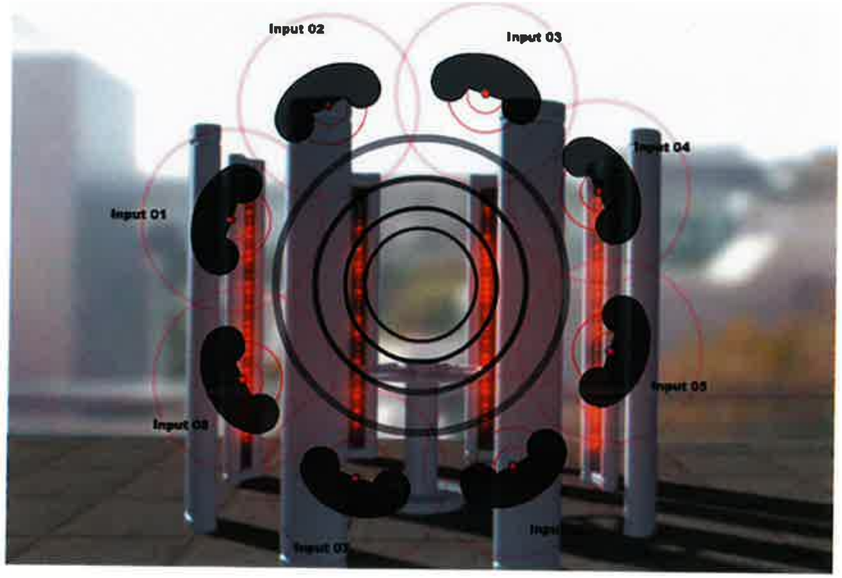
The participatory, networked creation of soundscapes
is also increasingly explored through the use of portable
'instruments'. Expanding his musical work into the realm
of nomadic devices, Golan Levin (with nine collaborators)
created *Telesymphony* (2001), a performance where sounds
were generated by the choreographed ringing of the
audience's mobile phones. The concert took place at the Ars
131 Electronica Festival in Linz in 2001. Audience members were
asked to register their phone number at a webkiosk before
the event and in return received a ticket which assigned them
a seat in the concert hall. New ring tones were automatically
downloaded to their mobile phones. Since each spectator at
the concert had a registered phone number, seat, and ring
tone, the performance itself could be precisely choreographed
by the musicians/performers. The audience thus became a
distributed melody in a 'cellular' space, while the disruption
often induced by the ring of a phone was unified into a
symphony. An installation-based project of a similar kind,
Telephony (2001) by Thomson & Craighead, allowed visitors

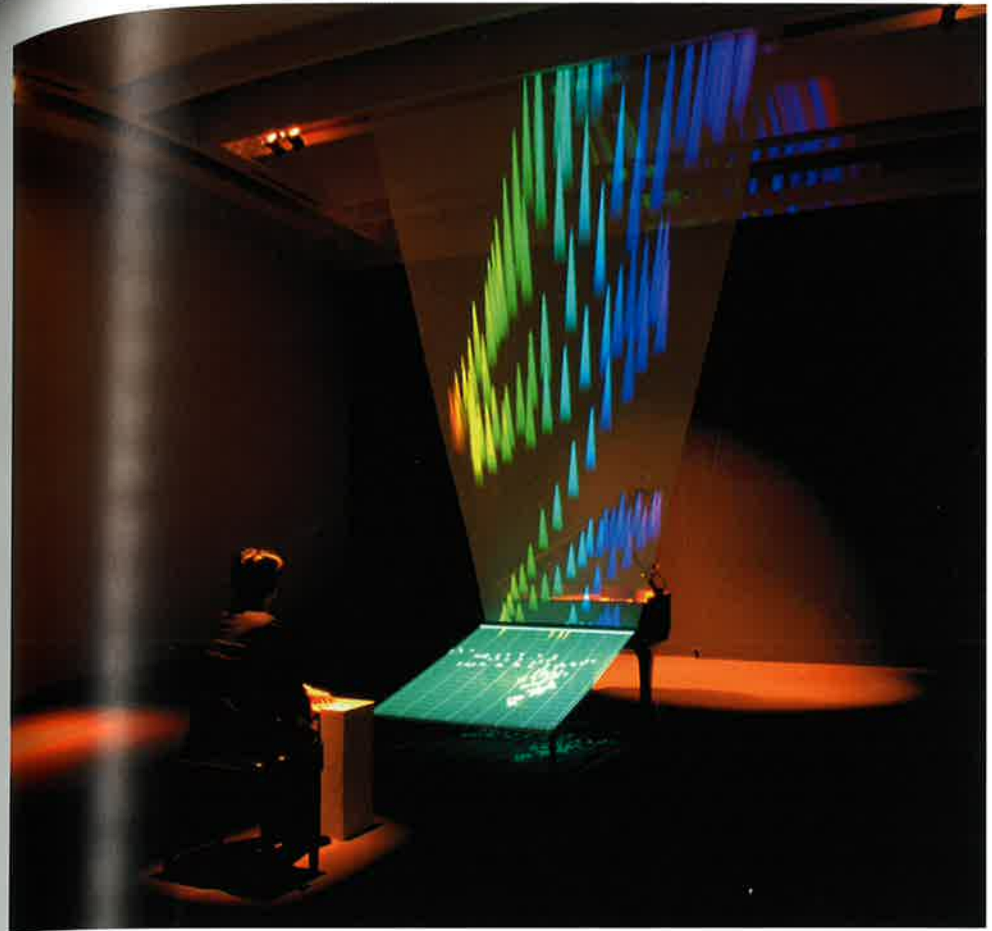
131 Golan Levin, *Telesymphony*, 2001. When an audience member's phone was 'played', a spotlight above them was switched on. The resulting grid of lights illuminating the audience was visible as a 'score' on projection screens at the sides of the stage. The performers dialled the phones by means of custom-software, and at times, nearly two hundred mobile phones were ringing simultaneously.

to a gallery and remote participants to dial up forty-two mobile phones that were installed as a grid on a wall. The phones would in turn start to also call each other, creating a layered audio environment. Works such as *Telesymphony* and *Telephony* continue the explorations of pioneers such as Max Neuhaus, who defined new arenas for music performance by staging sound works in public arenas and experimenting with networked sound as a form of 'virtual architecture'. In the first instalment of his project *Public Supply* (1966), he established a connection between the WBAI radio station in New York and the telephone network, implementing a twenty-mile aural space around New York City, where participants could intervene in the performance by making a phone call.

Sound and music projects also commonly take the form of interactive installations or 'sculptures' that respond to different kinds of user input or translate data into sounds and
132 visuals. The sound 'sculpture' *Ping* (2001) by American Chris Chafe (b. 1952) and Swiss-born Greg Niemeyer (b. 1967) is an audio-networking project driven by data travelling over the Internet. The sound created by the installation is created by ping commands, which contact servers to see if a connection can be established and thus provide a form of measuring time and distance. *Ping* translates the time lag of the data flow into audible information. Through the installation, users can pick

132 Chris Chafe and Greg Niemeyer, *Ping*, 2001





133 Toshio Iwai, *Piano—as image media*, 1995

instruments and scales or influence speaker configurations, as well as add to or change the list of websites to be 'pinged'. The invisible phenomenon of the rhythm and time of Internet traffic is transformed into a sensory experience. In Japanese artist Toshio Iwai's (b. 1962) interactive audio-visual
133 installation *Piano—as image media* (1995), a virtual score is used to trigger the keys of a piano, which in turn induce the projection of computer-generated images on a screen. The score is 'written' by the visitors who can position dots on a moving grid projected in front of the piano. The melodies played, as well as the visuals they produce, are generated through the pattern of dots assembled by the users. Iwai's

project establishes connections between notation, sound and visuals, as well as between the mechanical and virtual. The piano as physical object manifests itself as 'image media', both driven by and initiating different media elements. Iwai's project underscores the fluidity and easy translatability of diverse media elements that are characteristic of the digital medium and produces a synaesthetic experience, abandoning the focus on any one single element of sensory experience.

The fusion of visuals and sound in an installation and
134 performative environment also unfolds in *Supralunar* (2018) by Colombian artist Juan Cortés (b. 1989), which creates a sonic and visual experience of discoveries made in the 1970s by astronomer Vera Rubin. Rubin proposed the existence of invisible dark matter that keeps galaxies from being torn apart by the speed at which they rotate. Taking the sculptural form of an ancient clock with electromechanical gears in motion, the work simulates dark matter and the formation of a galaxy through light and sound, inviting the audience to rethink human–machine relations. When the viewer places an eye against the lens to watch the light simulation, the skull's orbital and temporal bones become an amplifier for transmitting the sound generated by the electromechanical gears inside. Cortés and the collective Atractor Studio, which he co-founded, have also performed this visualization in the live audio-visual performance *Dark Matter* (2018–19), underscoring the porous boundaries between scientific theories and the humanities in their respective constructions of reality and truth.

Electromechanical components are frequently used in both digital and analogue sound art, which creates an overlap and mutual influence between artistic practices. Swiss artist Zimoun (b. 1977) has become known for creating large-scale sound installations—made from off-the-shelf, low-tech materials and powered by motors—that have been an inspiration for digital sound art. Zimoun's site-specific installation *240 prepared dc-motors, cardboard boxes*
135 *60x20x20 cm* (2015) consists of a room full of tall, standing cardboard boxes that move and shake driven by ominous forces, thereby creating a sound architecture. The rhythm of movement and sound suggests both a prepared, controlled system, and a living structure that evolves by chance, never completely revealing what produces it. The power and impact of the work resides in its sensitivity to location and its creation of a system of movement and sound that carefully balances order, chance, and variation, the mechanical and the natural, the pre-determined and dynamism.

134 Juan Cortés, *Supralunar*, 2018

135 Zimoun, *240 prepared dc-motors, cardboard boxes, 60x20x20cm*, 2015

The generative element in Zimoun's work is a key characteristic of the digital and crucial to digital music and sound art. Generative explorations can take a wide range of forms, from immersive audio environments to invisible, subtle interventions. The complex installation *Radiosands*
136 (2017–19) by artist Thom Kubli (b. 1969) reflects on the current media landscape, which inundates us with chunks of news and disinformation with ever increasing speed and quantity. Consisting of a room-size array of radios on wire stands that have a both futuristic and retro aesthetic, *Radiosands* synthesizes these media flows into a spatial acoustic continuum. The radios have been modified so that they can individually be controlled by a composition software, which takes live audio as source and distributes chunks of sound across the installation. The software uses a semantic speech analysis program to search different radio streams for terms related to the construction of social and political reality, as well as human perception. The installation highlights how

136 Thom Kubli, *Radiosands*, 2017–19

137 Richard Garet, *Quotidian*, 2017

media shape our beliefs and experience in an immersive sonic field. On the opposite end of generativity's expressive spectrum
137 would be *Quotidian* (2017) by Uruguayan artist Richard Garet (b. 1972), a generative quadraphonic sound installation that consists of four transducers hidden behind a wall. The work makes itself 'visible' to visitors only through an eight- to ten-foot wall drawing that outlines the outer limits of the space where listeners can place their ears to experience the work. Algorithmically generated in new ways each time it plays spatially across the four transducers, *Quotidian*'s composition comprises everyday sounds one encounters in common spaces and plays with our everyday routine of trying to identify them. As listeners press their ears towards the wall, they also construct its space by leaving their marks in the form of sweat, oil, or make-up.

138 Sara Ludy, *Deeptimesurfacetime*, 2019

Digital sound art frequently reflects on its own inherent characteristics, sometimes by juxtaposing the sonic qualities of naturally and algorithmically generated audio spaces.
138 Sara Ludy's (b. 1980) *Deeptimesurfacetime* (2019) is a roughly ten-minute sound piece that fuses 'field recordings' from two environments: Nan Lian Garden in Hong Kong, a highly designed monastery garden with hills, rock structures, bridges, pavilions and ponds, and a virtual aviary that Ludy built in VR. The mix of natural and simulated digital sounds creates a sonic landscape that questions and plays with constructs of nature.